

CS-1004
Object Oriented Programming
Assignment 03

Objectives:

-  Classes
-  Objects
-  Constructors

Total Questions: 04

Total Marks: 80

Deadline : 26th October 2023

4 : 30 PM

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. There must be a block of comments at the start of every question's code by students; the block should contain brief description about functionality of code.
2. Comment on every function and about its functionality.
3. Mention comments where necessary such as comments with variables, loop, classes etc. to increase code understandability.
4. Use understandable names of variables.
5. Proper indentation of code is essential.
6. Write a code in C++ language.
7. Make a Microsoft Word file and paste all your C++ code with all possible screenshots of every task **output in Microsoft Word and submit the word file as well.**
8. First think about statement problems and then write/draw your logic on copy. 9. After copy pencil work, code the problem statement on MS Studio C++ compiler. 10. At the end when you have done your tasks, attached C++ created files in MS word file and make your submission on Google Classroom. (Make sure your submission is completed).
9. Please submit your file in this format **22F_XXXX_A1_Sec.zip**.
10. **Do not submit your assignment after the deadline. Late submission is not accepted.**
11. **Do not copy code from any source otherwise you will be penalized with negative marks or zero.**

Question No. 1

(For Q1, Please refer to the folder provided to you in order to complete the code)

For this challenge you are to develop the foundation for a program for movie fanatics to keep track of what movies they have watched and how many times they watched each movie.

The program must support the following:

class Movie - models a movie which includes

- movie name
- movie rating (G, PG, PG-13, R)
- watched - the number of times the movie has been watched

class Movies - models a collection of movie objects

Obviously, Movies needs to know about Movie since it is a collection of Movie object. However, our main driver should only interact with the Movies class.

Q2-

Implementation of String Class: Your goal is to implement a generic 'String' class using cstring. Your implemented class must fully provide the definitions of following class (interface) functions. Please also write down the test code to drive your class implementation. **Please note that we will be running your code against our test code and any segmentation faults or incorrect result will result in loss of marks.**

```
1 class String{
2 // think about the private data members...
3 public:
4 // provide definitions of following functions...
5     String();// default constructor
6     String(char *str);// initializes the string with constant cstring
7     String(const String &);// copy constructor to initialize the string from
        ↳ existing string
8     String(int x);// initializes a string of pre-defined size
9     char getAt(int i);// returns the character at index [x]
10    void setAt(int i, char c);// set the character at index [x]
11    String substr(int pos, int len);// returns a substring of length len starting
        ↳ from location 'pos'
12    String substr(int pos);// returns substring from the given position to the end.
13    void append(char a);// append a char at the end of string
14    void append(String str );// append a String at the end of string
15    void append(char *str );// append a constant c string at the end of string
16    int length();// returns the length of string
17    char* tocstring();// converts a String to c-string
18    void display();// displays the string ..
19    bool isEmpty();// returns true if string is empty..
20    void copy(const String&);// Copy one string to another ...
21    void copy(const char *);// copy cstring to String...
22    int find(char);// returns the index of character being searched.
23    bool equal(String);// should return true if both strings are same
24    int stoi();// function for converting a string to integer.
25    void split(char token, string *&, int &ntokens); // should split the string
        ↳ according to given token and store all substrings in a dynamic array
        ↳ passed as second argument, and return ntokens in third argument.
26    bool isanagram(String &);// return true if the given string is anagram of the
        ↳ input string, an string is called anagram of another string if it contains
        ↳ exactly same characters but might be different order of appearance.
27    ~String();// destructor...
28 };
```

Q3: Implementation of Array Class: Your goal is to implement an integer 'Array' class. Your implemented class must fully provide the definitions of following functions. Please also write down the test code to drive your class implementation.

```

1 class Array{
2   // think about the private data members...
3 public:
4   // provide definitions of following functions...
5   Array(); // a default constructor
6   Array(int size); // a parametrized constructor initializing an Array of
    ↳ predefined size
7   Array(int *arr, int size); // initializes the Array with an existing Array
8   Array(const Array &); // copy constructor
9   int getAt(int i); // returns the integer at index [i]
10  void setAt(int i, int val); // set the value at index [i]
11  Array subArr(int pos, int siz); // returns a sub-Array of size siz starting from
    ↳ location 'pos'
12  Array subArr(int pos); // returns a sub-Array from the given position to the end.
13  int* subArr(int pos, int siz); // returns an array of size siz starting from
    ↳ location 'pos'
14  int* subArr(int pos); // returns an array from the given position to the end.
15  void push_back(int a); // adds an element to the end of the array
16  int pop_back(); // removes and returns the last element of the array
17  int insert(int idx, int val); // inserts the value val at idx. Returns 1 for a
    ↳ successful insertion and -1 if idx does not exists or is
    ↳ invalid. Shift the elements after idx to the right.
18  int erase(int idx, int val); // erases the value val at idx. Returns 1 for a
    ↳ successful deletion and -1 if idx does not exists or is
    ↳ invalid. Shift the elements after idx to the left.
19  int length(); // returns the size of the Array
20  void clear(); // clears the contents of the Array
21  int value(int idx); // returns the value at idx
22  void assign(int idx, int val); // assigns the value val to the element at index
    ↳ idx
23  void copy(const Array& Arr); // Copy the passed Array
24  void copy(const int * arr, int siz); // copy the passed array
25  void display(); // displays the Array
26  bool isEmpty(); // returns true if the Array is empty
27  Array find(int); // returns an Array containing all the indexes of integer being
    ↳ searched
28  bool equal(Array); // should return true if both Arrays are same
29  int sort(); // sorts the Array. Returns true if the array is already sorted
30  void reverse(); // reverses the contents of the array
31  ~Array(); // destructor...
32 };
  
```

Q4: Implementation of Matrix Class: In this last question you have to implement the 'Matrix' class by defining the following functions. Please also write down the test code to drive your class implementation

```
1 class Matrix{
2 // think about the private data members...
3 // the matrix should store real numbers
4 public:
5 //include all the necessary checks before performing the operations in the functions
6 Matrix(); // a default constructor
7 Matrix(int, int); // a parametrized constructor
8 Matrix(const Matrix &); // copy constructor
9 void set(int i, int j); //set value at (i,j)
10 float get(int i, int j); //get value at (i,j)
11 Matrix assign(const Matrix &); //assigns (copies) a Matrix. Returns the same
12 Matrix add(const Matrix &); //adds two Matrices and returns the result
13 Matrix subtract(const Matrix &); //subtracts two Matrices and returns the result
14 Matrix multiply(const Matrix &); //multiplies two Matrices and returns the result
15 Matrix add(float); //assigns a constant to every element
16 Matrix multiply(float); //multiplies every element with a constant
17 ~Matrix();
```